Arcs complete outside a prescribed conic

Exact defect, equality rigidity, and $\rho_{\mathcal{C}}(16) = 9$

Tavis Rudd

July 2026

Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2, q)$. We study arcs $A \subseteq \mathrm{PG}(2, q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Starting from the classical first two equations for the secant-index distribution of an arc, we prove an exact defect identity for an arbitrary prescribed set of points that may remain uncovered. Its nonnegative local remainder gives exact equality and quantitative stability criteria. For a conic, writing $I_{\mathcal{C}}(A)$ for the total incidence of the secants of A with $\mathcal{C}$, it gives

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{\lfloor k/2 \rfloor}$$

for every $\mathcal{C}$-complete k-arc, together with an exact equality criterion and, for every prime power q,

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

At equality, the concurrence points of disjoint secants form a matching design: they partition the edges of the Kneser graph $KG(k, 2)$ into cliques of maximum order. This gives a global rigidity theorem and a quantitative graph-theoretic form of stability. We also prove an upper transfer by projective averaging. Explicit witnesses meet the lower bound for $q = 5, 8, 9, 11$; at $q = 5$ the witness is a projective frame. Over $\mathbb{F}_{16}$, a certificate-checked exhaustive covering list shows that no nonzero quadratic contains an eight-arc's ordinary-uncovered locus while avoiding the arc; hence $\rho_{\mathcal{C}}(16) = 9$.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [5], Kim–Vu [15], and Nagy [20].

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \mathrm{PG}(2, q) \setminus \mathcal{C}$ is $\mathcal{C}$-*complete* if every point of

$$\mathrm{PG}(2, q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

More generally, if $\mathcal{H}$ is any set disjoint from an arc A, write

$$\mathcal{U}_{\mathcal{H}}(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \mathcal{H} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

We abbreviate $U(A) = \mathcal{U}_{\varnothing}(A)$ for the ordinary-uncovered locus and retain $\mathcal{U}_{\mathcal{C}}(A)$ for the conic-relative locus. Thus A is $\mathcal{C}$-complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$.

The minimum is attained because a maximum-cardinality arc disjoint from $\mathcal{C}$ is maximal under insertion. Equivalently, A is maximal among arcs in the point set $\mathrm{PG}(2,q) \setminus \mathcal{C}$. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

The problem differs from two nearby standard notions. A saturating set need not be an arc and has no prescribed exceptional locus. An almost-complete subset of a conic, in the sense of Bartoli–Davydov–Marcugini–Pambianco [7], consists of selected points *on* the conic whose chords cover the complementary points. Ng–Wild's arcs covering a line [21] prescribe a one-dimensional set that must be covered, rather than an exceptional set that may remain uncovered. Here the selected arc is required to avoid the conic.

The arbitrary-hole framework also contains an established line-hole problem. If the prescribed set is a line L_∞ and the arc avoids that line, completeness outside L_∞ is exactly completeness as an arc in the affine plane $\mathrm{PG}(2,q) \setminus L_\infty$. Closely related hyperfocused arcs were motivated by geometric secret sharing [13], and some constructions are complete in precisely this off-line sense [13, 16]. Corollary 3.8 below gives the resulting affine defect inequality and exact equality criterion. The conic problem is therefore a curved exceptional-locus specialization, rather than the sole reason for the prescribed-hole formulation.

Localization of an uncovered locus on a proper subgeometry is also known for curve-derived arcs with larger line intersections. Korchmáros, Nagy, and Szőnyi [17, Theorem 7.5] prove that, for odd q and odd $r \geq 5$, the points left uncovered by the $(q+1)$-secants of their rational BKS $(k, q+1)$-arc in $\mathrm{PG}(2, q^r)$ are exactly the points of the proper subplane $\mathrm{PG}(2,q)$ not belonging to the arc. Thus localization of holes in a prescribed subplane is prior art. Their theorem concerns a particular curve-derived $(k, q+1)$-arc; the contribution here is instead the exact defect identity for an arbitrary prescribed hole set, together with its equality and stability theory and its specialization to a prescribed conic for ordinary arcs.

There is also classical work on *complete exterior sets*: sets of $(q+1)/2$ exterior points whose pairwise joins miss the conic [8, p. 143]. This condition constrains the joins of the selected points, whereas Definition 1.1 requires those joins to cover the complementary off-conic locus. The two conditions therefore point in opposite directions:

$$\begin{aligned} \text{complete exteriority} \quad &\Longrightarrow \quad \mathcal{C}(\mathbb{F}_q) \subseteq U(A), \\ \mathcal{C}\text{-completeness} \quad &\Longleftrightarrow \quad U(A) \subseteq \mathcal{C}(\mathbb{F}_q), \\ \text{both conditions} \quad &\Longrightarrow \quad U(A) = \mathcal{C}(\mathbb{F}_q). \end{aligned}$$

Strict containment at $q = 7$ and the exceptional equality at $q = 11$ return in Section 8.1.

There is also a standard coding formulation. For a candidate of size $k \geq 4$, the arc–MDS dictionary identifies its columns with a projective $[k, k-3, 4]_q$ parity-check system. Under this dictionary, $\mathcal{C}$-completeness says exactly that every projective syndrome direction whose cosets have distance three is confined to the prescribed conic; see Proposition 8.1. Thus $\rho_{\mathcal{C}}(q)$ is also a minimum-length problem for projective codimension-three MDS systems with a prescribed algebraic locus allowed to contain their distance-three syndrome directions.

The point index with respect to an arc, and the first two double-counting equations for its distribution, are classical; compare Hirschfeld [14, Chapter 9] and Ball [5]. The use of secant counts to bound the smallest complete (k, n)-arcs is also classical; for a recent explicit lower bound see Alabdullah–Hirschfeld [2]. The main observation is that, after splitting those equations over a prescribed exceptional set and its complement, the inequalities normally used to control overlap have an exact remainder. This defect identity is the principal result: it turns the usual lower-bound argument into a local accounting theorem from which equality, stability, and the later coding interpretation all follow. Its main consequences and applications are:

(i) an exact prescribed-hole defect identity, with equality and stability;

(ii) a matching-design classification of the global equality structure;

(iii) conic lower bounds, including the additive asymptotic term $3/2$;

(iv) a projective-averaging upper-bound transfer; and

(v) exact values at $q = 5, 8, 9, 11$, and a certificate-checked quadratic obstruction giving $\rho_{\mathcal{C}}(16) = 9$.

The equality structure belongs to the established theory of matching designs. Alspach and Heinrich [4] call a family of s-edge matchings in K_n a $\mathrm{MATCH}(n, s, \lambda)$ design when every pair of independent edges occurs in exactly λ members. Our contribution at this interface is the geometric implication: vanishing of the prescribed-hole defect canonically produces a simple $\mathrm{MATCH}(k, \lfloor k/2 \rfloor, 1)$ design, represented by concurrence points of secants in one projective plane. The local remainder also measures the graph edges that fail to lie in such maximum-matching cliques.

Sections 2–7 form the geometric route: exact accounting, lower and upper bounds, the characteristic-two refinement, and the finite obstruction. Section 8 then gives a secondary coding reading, the $q = 11$ application, and a fixed-length reconstruction theorem showing when the complete uncovered locus determines its parent arc; it may be skipped without affecting the geometric argument.

The secant-index equations, the Lunelli–Sce $\sqrt{2q}$ scale, the arc–MDS correspondence, and the Clebsch geometry at $q = 11$ are classical inputs. We claim neither those ingredients nor the known ordinary eight-arc class count. The contributions are the prescribed-hole remainder with its equality and stability consequences, its explicit conic specialization, the uncovered-locus evaluation obstruction, and the certified relative-conic conclusions. The $q = 11$ deep-hole equality is an apparently unrecorded synthesis of classical geometry with the relative-completeness and coding dictionaries, not a claim of priority for the Clebsch configuration itself.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q, and let A be a k-arc in Π. Write $r_A(x)$ for the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A. Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A, of which there are at most $\lfloor k/2 \rfloor$. $\square$

Proposition 2.2 (Classical first and second index equations)**.** *For every k-arc A in a projective plane of order q,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A, and therefore contributes $q-1$ incidences with points outside A. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. $\square$

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered. The notation used in the identity is summarized by

$V_{\mathcal{H}}(A) = \Pi \setminus (A \cup \mathcal{H})$	required locus
$\mathcal{X}_{\mathcal{H}}(A) = \{x \in V_{\mathcal{H}}(A) : r(x) > 0\}$	covered required points
$\mathcal{U}_{\mathcal{H}}(A) = V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A)$	uncovered required points
$I_{\mathcal{H}}(A) = \sum_{y \in \mathcal{H}} r(y)$	secant incidence spent on the holes.

The third line agrees with the uncovered-locus notation introduced in Definition 1.1; in particular, $U = \mathcal{U}_{\varnothing}$ and the conic-relative locus is $\mathcal{U}_{\mathcal{C}}$.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

Theorem 3.1 (Prescribed-hole defect identity)**.** *Let A be a k-arc in a projective plane of order q, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

It is the exact coverage capacity left after the universal second-moment and hole-incidence losses; the identity below resolves that capacity into local nonextremal-index contributions. Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y\in\mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y\in\mathcal{H}} \binom{r(y)}{2}.$$

Substitution first separates the required-point and hole terms:

$$\begin{aligned} m\Delta_{\mathcal{H}}(A) = &\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \left(m(r(x)-1) - 2\binom{r(x)}{2} \right) \\ &+ \sum_{y\in\mathcal{H}} \left((m-1)r(y) - 2\binom{r(y)}{2} \right). \end{aligned}$$

The pointwise identities

$$\begin{aligned} m(r-1) - 2\binom{r}{2} &= (r-1)(m-r), \\ (m-1)r - 2\binom{r}{2} &= r(m-r), \end{aligned}$$

now give (5). Every term is nonnegative by Lemma 2.1. □

Corollary 3.2 (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \leq N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \geq |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \geq 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). □

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes)**.** *Let $|\mathcal{H}| = h$. If A is a k-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2+q+1-k-h$. Substitute this equality into (6). $\square$

The exact remainder also quantifies near equality.

Corollary 3.4 (Quantitative stability)**.** *Assume $m \ge 3$ and put*

$$\begin{aligned} M &= \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \le r(x) \le m-1\}, \\ J &= \{y \in \mathcal{H} : 1 \le r(y) \le m-1\}. \end{aligned}$$

Then

$$(m-2)|M| + (m-1)|J| \le m\Delta_{\mathcal{H}}(A). \tag{9}$$

Proof. For $2 \le r \le m-1$, $(r-1)(m-r) \ge m-2$. For $1 \le r \le m-1$, $r(m-r) \ge m-1$. Apply these estimates to (5). $\square$

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m, while almost every hole point met by a secant also has multiplicity m. This is the precise stability content behind the equality heuristic.

3.1 Concurrency designs at equality

The equality criterion has a global incidence interpretation. Label the vertices of K_k by the points of A, so that its edges label the secants of A. Two such edges are adjacent in the Kneser graph $KG(k,2)$ exactly when the corresponding secants have disjoint endpoint pairs.

Theorem 3.5 (Concurrency decomposition and equality rigidity)**.** *Suppose $k \ge 4$. For each $x \notin A$ with $r(x) \ge 2$, the secants through x form an $r(x)$-edge matching M_x in K_k, and*

$$E(KG(k,2)) = \dot{\bigcup_{x\notin A,\ r(x)\ge 2}} E(K_{M_x}).$$

Here K_{M_x} is the clique whose vertices are the edges of M_x.

If $\Delta_{\mathcal{H}}(A) = 0$, every M_x in this decomposition has $m = \lfloor k/2 \rfloor$ edges. Thus the family $(M_x)_{r(x)=m}$ is a simple MATCH$(k,m,1)$ *design. If $Z = \{x \notin A : r(x) = m\}$, then*

$$|Z| = \frac{3\binom{k}{4}}{\binom{m}{2}} = \begin{cases} (k-1)(k-3), & k \text{ even}, \\ k(k-2), & k \text{ odd}, \end{cases}$$

and every secant of A contains $k-3$ points of Z when k is even and $k-2$ points of Z when k is odd.

Proof. Secants through a point outside an arc have pairwise disjoint endpoint pairs, so they form a matching. Conversely, two secants with disjoint endpoint pairs meet in a unique point outside A. Their edge in $KG(k,2)$ therefore belongs to exactly one concurrence clique, proving the decomposition.

If the defect vanishes, Theorem 3.1 gives $r(x) \in \{1, m\}$ off $\mathcal{H}$ and $r(x) \in \{0, m\}$ on $\mathcal{H}$. Every concurrence point has index at least two, so all the cliques above have order m. The second index equation now gives $|Z|\binom{m}{2} = 3\binom{k}{4}$.

Fix a secant. It is disjoint from $\binom{k-2}{2}$ other secants, and each point of Z on it accounts for $m-1$ of them. Hence it contains $\binom{k-2}{2}/(m-1)$ points of Z, which has the two displayed values. □

After any projective duality, the $\binom{k}{2}$ secants become points. They lie on k star lines of size $k-1$, one for each point of A, and on the $|Z|$ matching lines of size m. Two dual secant-points lie on a star line when their endpoint pairs meet and on a unique matching line when they are disjoint. Thus zero defect does not merely prescribe the indices: it gives a rank-three projective realization of the star–matching pairwise-balanced design on the edges of K_k.

The same decomposition strengthens Corollary 3.4 from a count of nonextremal centres to a count of affected pairs of secants.

Corollary 3.6 (Bad-edge stability). *Let E_{bad} be the set of edges of $KG(k,2)$ lying in a concurrence clique of order $2 \le r(x) \le m-1$. Then*

$$|E_{\mathrm{bad}}| \le \frac{m(m-1)}{2}\,\Delta_{\mathcal{H}}(A).$$

Proof. At an off-hole centre of index r,

$$\binom{r}{2} \le \frac{m-1}{2}(r-1)(m-r),$$

and at a hole centre the defect weight $r(m-r)$ is larger. Sum these inequalities and apply (5). □

The first two nontrivial orders illustrate the distinction between an abstract matching design and its rank-three projective realization.

Corollary 3.7 (Six and seven points). *(a) In a Desarguesian plane, if a six-arc has every intersection of two disjoint secants on a third secant, then the field has characteristic two, contains $\mathbb{F}_4$, and the arc is projectively equivalent to*

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1),\ (1,\omega,\omega^2),\ (1,\omega^2,\omega), \qquad \omega^2+\omega+1=0.$$

Conversely, this configuration has the stated concurrence property. In particular, every zero-defect six-arc in a Desarguesian plane has this form.

(b) In any projective plane, no seven-arc has zero defect with respect to any prescribed hole set.

Proof. For six points, the three intersections of opposite sides of any complete quadrangle must all lie on the secant through the remaining pair. Normalize four vertices to

$$(1,0,0), \quad (0,1,0), \quad (0,0,1), \quad (1,1,1).$$

Their diagonal points are collinear only in characteristic two; their common line is then $x+y+z=0$, and it contains the remaining pair. Write a fifth point as $(1,t,1+t)$. Applying the same diagonal-line condition to the first three frame points and this fifth point gives $t^2+t+1=0$; the sixth point is then $(1,t^2,t)$. Direct substitution shows that the displayed configuration has the required concurrences.

For seven points, Theorem 3.5 would produce a MATCH$(7,3,1)$ design. Alspach and Heinrich [4, Theorem 3.1] proved that no such design exists. □

The line-hole case now makes the connection with complete affine arcs explicit.

Corollary 3.8 (Complete affine arcs)**.** *Let L_∞ be a line in a projective plane of order q, and let A be a k-arc disjoint from L_∞. Then A is complete as an arc in the affine plane $\Pi \setminus L_\infty$ if and only if it is complete outside L_∞. In that case*

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}\binom{k}{2}. \tag{10}$$

Equality holds if and only if every affine point outside A has secant index in $\{1, m\}$ and every point of L_∞ has secant index in $\{0, m\}$.

Proof. Maximality in the affine plane is equivalent to every affine point outside A lying on a secant of A, which is exactly completeness outside L_∞. Every secant meets L_∞ once, so $I_{L_\infty}(A) = \binom{k}{2}$. Now apply Corollaries 3.3 and 3.2; the latter also gives the equality criterion. □

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$, and let $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ be a k-arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 4.1 (Conic-loss and uncovered-locus inequalities)**.** *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \ge q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{11}$$

If A is $\mathcal{C}$-complete, then

$$\boxed{q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{12}$$

Proof. Apply Corollary 3.2 with $\mathcal{H} = \mathcal{C}$, using $|V_{\mathcal{C}}(A)| = q^2 - k$ and $I_{\mathcal{H}}(A) = I_{\mathcal{C}}(A)$. If A is $\mathcal{C}$-complete, then $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$, and the first inequality rearranges to (12). □

The first correction in (12) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Define

$$L_1(q) = \min\left\{k \ge 3 : q^2 - k \le \binom{k}{2}(q-1)\right\},$$
$$L_2(q) = \min\left\{k \ge 3 : q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4}\right\}.$$

Then

$$\rho_{\mathcal{C}}(q) \ge L_2(q) \ge L_1(q). \tag{13}$$

Indeed, for $q \ge 3$ a set of at most two points has at most one secant, and that secant together with the set and the conic contains at most $2q + 2 < q^2 + q + 1$ points. Thus a $\mathcal{C}$-complete arc has at least three points, so the moment bound applies to the unrestricted minimum defining $\rho_{\mathcal{C}}(q)$. The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$.

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} = \begin{cases} \dfrac{k-1}{2}\big(k(q-1) - (k-2)(k-3)\big), & k \text{ even}, \\ \dfrac{k}{2}\big((k-1)(q-1) - (k-2)(k-3)\big), & k \text{ odd}. \end{cases} \tag{14}$$

For comparison, some values are

q	5	8	9	11	13	16	17	19
$L_1(q)$	4	5	5	6	6	7	7	7
$L_2(q)$	4	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2,16)$ has at least eight points.

The routine algebra behind the additive bound is isolated in the next lemma on the full interval needed to avoid any parity split.

Lemma 4.2 (Polynomial estimate)**.** *For $s \geq 2$ and $0 \leq a \leq 2$, put*

$$P_s(a) = \frac{2a-3}{4}s^3 + \frac{a^2-7a+10}{4}s^2 + \frac{-3a^2+10a-8}{2}s + \frac{-a^3+5a^2-8a+6}{2}.$$

If $P_s(a) \geq 0$, then $a \geq 3/2 - 8/s$.

Proof. On $0 \leq a \leq 2$, the coefficients of s^2, s, and 1 in $P_s(a)$ are at most $5/2$, 1, and 3, respectively. Since $s \geq 2$, their total contribution is at most

$$\frac{5}{2}s^2 + s + 3 \leq 4s^2.$$

Thus $P_s(a) \geq 0$ implies $0 \leq (2a-3)s^3/4 + 4s^2$, and division by s^2 gives the claim. □

Theorem 4.3 (Explicit additive lower bound)**.** *For every prime power q,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{15}$$

Consequently, as q tends to infinity through prime powers,

$$\liminf_{q\to\infty}\big(\rho_{\mathcal{C}}(q) - \sqrt{2q}\big) \geq \frac{3}{2}. \tag{16}$$

Proof. Let k be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (15) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m}\binom{k}{4} \geq \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (12) therefore gives the necessary inequality

$$q^2 - k \leq \frac{k-1}{2}\big(k(q-1) - (k-2)(k-3)\big).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is $P_s(a)$. It is nonnegative, so Lemma 4.2 gives $a \geq 3/2 - 8/s$, which is (15). Letting q tend to infinity through prime powers gives (16). □

Remark 4.4 (The scale of the conic term). Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \le I_{\mathcal{C}}(A) \le 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (12) can be useful for finely balanced finite cases, but cannot improve the additive constant 3/2 in Theorem 4.3. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

5 An upper-bound transfer from complete arcs

Let $t_2(2, q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2, q)$.

Theorem 5.1 (Projective averaging)**.** *Let $H \subseteq \mathrm{PG}(2, q)$ and let B be an ordinary complete b-arc. If*

$$b|H| < q^2 + q + 1, \tag{17}$$

then some projective image of B is disjoint from H and complete outside H.

Proof. Choose a uniformly random projectivity $g \in \mathrm{PGL}(3, q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2 + q + 1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g. Projectivities preserve ordinary completeness, so gB is complete outside H. □

Corollary 5.2 (Conic transfer)**.** *If $t_2(2, q) \le q$, then*

$$\rho_{\mathcal{C}}(q) \le t_2(2, q). \tag{18}$$

More generally, every complete b-arc with $b \le q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q + 1$. For integral $b \le q$ one has $b(q + 1) < q^2 + q + 1$, and Theorem 5.1 applies. □

Kim and Vu proved that, for all sufficiently large q, every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [15]. This is less than q for large q, so Corollary 5.2 gives the following immediate consequence.

Corollary 5.3. *For some absolute constant c and all sufficiently large prime powers q,*

$$\rho_{\mathcal{C}}(q) \le \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

6 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν, and every line through ν is one of these $q+1$ tangents.

Proposition 6.1 (Case $\nu \in A$)**.** *Let A be a k-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ A-secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \geq k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

If A is $\mathcal{C}$-complete, then

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{19}$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k-1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν, which would place three points of A on one line. Hence these are exactly the tangent A-secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (19) follows from Corollary 4.1. $\square$

Proposition 6.2 (Case $\nu \notin A$)**.** *Let A be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

the tangent A-secants are exactly the secants through ν, and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

Consequently,

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{20}$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν. Thus the number of tangent A-secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 4.1. $\square$

Corollary 6.3 (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

Consequently, inequality (12) *can always be strengthened by replacing its final term by at least $-1/m$.*

Proof. If the nucleus belongs to A, Proposition 6.1 gives $I_{\mathcal{C}}(A) \geq k-1 \geq 1$. Otherwise Proposition 6.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. $\square$

These two cases are useful structural reductions, but Remark 4.4 limits what the incidence term alone can accomplish. For example, at $(q,k) = (16,8)$ inequality (12) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

7 Verified finite-field examples

The verified values at $q = 5, 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 5$ the witness is the projective four-frame. At $q = 16$, the same ingredients give $8 \leq \rho_{\mathcal{C}}(16) \leq 9$; a checked projective classification excludes eight. The proof has three steps: the defect bound supplies the lower bound eight; a hypothetical $\mathcal{C}$-complete eight-arc would have its ordinary-uncovered locus on a quadratic avoiding the arc; and the certificate rules out every such quadratic, while the displayed nine-point witness supplies the upper bound.

7.1 Evaluation obstruction

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [12] and Ball [6]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of x. The zero or nonzero status of this evaluation is independent of the chosen representative.

Indeed, if an arc is complete outside a conic, then its ordinary-uncovered locus lies in the zero set of the conic equation while every selected point avoids that zero set. The next lemma isolates exactly the linear-algebraic obstruction used to rule this out at $q = 16$.

Lemma 7.1 (Uncovered evaluation obstruction)**.** *Let A and U be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*
2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

Proof. If f vanishes on U, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$. □

For the full system of degree-d forms, the evaluation functionals are the degree-d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

The more general problem of avoiding several selected points at once has a sharp finite-field answer, recorded in Appendix B.

7.2 Small values and the classification over $\mathbb{F}_{16}$

Proposition 7.2. *For the standard conic*

$$\mathcal{C} : \; XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(5) = 4, \qquad \rho_{\mathcal{C}}(8) = 6, \qquad \rho_{\mathcal{C}}(9) = 6, \qquad \rho_{\mathcal{C}}(11) = 6.$$

Proof. Equation (13) gives

$$L_2(5) = 4, \qquad L_2(8) = L_2(9) = L_2(11) = 6.$$

For $q = 5$, the standard four-frame has ordinary-uncovered locus

$$V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. An invertible coordinate change carries this conic to $XZ = Y^2$ and the frame to the four coordinates in Appendix C; the relative certificate is checked in Lean. The appendix also lists $\mathcal{C}$-complete arcs of size six for $q = 8, 9, 11$. Their verification is described below. $\square$

For context, the companion treatment classifies all conic-filling loci for $4 \le k \le 7$ [1, the conic-filling classification]. That broader classification is not used in Proposition 7.2.

The ordinary classification of eight-arcs over $\mathbb{F}_{16}$ records arc counts, stabilizers, secant-index distributions, and conic-contained classes. The theorem below adds the datum needed here: the ordinary uncovered locus either imposes six independent quadratic conditions or forces every containing quadratic to meet the arc.

The argument needs only an exhaustive *covering list*, not a classification into pairwise inequivalent representatives. Lean kernel-checks every local transition, every leaf obstruction, frame normalization, and the global arbitrary-quadratic consequence. The C++ program generates certificate data but is not trusted for exhaustiveness or rejection; the known class count is used only as an external consistency check. Appendix A gives the exact theorem and regeneration boundary.

We first state the completeness contract used by the finite certificate. Index the 273 projective points by their unique normalized representatives $[0 : 0 : 1]$, $[0 : 1 : z]$, and $[1 : y : z]$. The standard four-frame is

$$F = \{[0 : 0 : 1], [0 : 1 : 0], [1 : 0 : 0], [1 : 1 : 1]\}.$$

Lemma 7.3 (Augmentation certificate contract)**.** *For $4 \le i \le 8$, let $\mathcal{L}_i$ be a finite list of i-arcs with $\mathcal{L}_4 = \{F\}$. Suppose that for every $4 \le i < 8$, every $P \in \mathcal{L}_i$, and every $x \notin P$ such that $P \cup \{x\}$ is an arc, the certificate supplies a child $P' \in \mathcal{L}_{i+1}$ and an invertible linear map whose induced projectivity carries $P \cup \{x\}$ onto P'. Then every eight-arc in* $\mathrm{PG}(2, 16)$ *is projectively equivalent to a member of $\mathcal{L}_8$.*

Proof. Choose four points of the eight-arc. Every four-arc is a projective frame, so a projectivity carries those points to F. Insert the remaining four points in any order. At each step the stated certificate transports the legal insertion to the next list. Induction through the four layers gives a member of $\mathcal{L}_8$ projectively equivalent to the original arc. $\square$

Theorem 7.4 (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let A be any eight-arc in* $\mathrm{PG}(2, 16)$*, and let $U(A)$ be the points outside A that lie on no secant of A. There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \qquad \textit{and} \qquad A \cap Z(Q) = \varnothing.$$

The assertion includes singular quadratic forms.

Proof. Certified covering lists. Starting from F, the checked list lengths are $|\mathcal{L}_5| = 4$, $|\mathcal{L}_6| = 61$, $|\mathcal{L}_7| = 454$, and $|\mathcal{L}_8| = 2633$. For every listed parent, the checker tests all 273 indexed points. Each legal move carries a child index, an invertible 3×3 matrix, and pointwise nonzero scalar witnesses proving that the induced projectivity maps the enlarged parent onto that child. A separate check matches the certificate books with all parents in the current list. Lemma 7.3 therefore proves exhaustiveness without trusting the generator's canonical labels or deduplication. As an external consistency check, the final list length agrees with the projective-class count of Al-Seraji–Al-Ogali [3, Theorem 3.8.1].

Leaf obstruction. For 2630 listed leaves, the certificate gives six ordinarily uncovered points and an explicit inverse to their 6×6 quadratic evaluation matrix. This is the first alternative of Lemma 7.1. For each remaining leaf, a checked linear combination expresses evaluation at a selected arc point in the span of evaluations at ordinarily uncovered points, giving the second alternative. Thus every listed leaf has the asserted quadratic-avoidance property.

Projective transport. If gA is a listed representative, replace Q by $Q \circ g^{-1}$. This form is nonzero, its zero set is $gZ(Q)$, and projectivities carry secants to secants, so $U(gA) = gU(A)$. A quadratic as in the statement would therefore contradict the checked obstruction for the listed leaf. Appendix A records the independent regeneration and formal trust boundary. □

Corollary 7.5 (Exact relative value over $\mathbb{F}_{16}$)**.** *For every nonsingular conic $\mathcal{C} \subseteq \mathrm{PG}(2,16)$,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (13) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem 7.4. The nine-point witness in Appendix C gives the matching upper bound. □

Remark 7.6 (Scope of the classification certificate)*.* The proof requires an exhaustive covering list, not pairwise inequivalence of its entries. Pairwise deduplication and the exact quotient-class count belong to the cited ordinary classification; their agreement with the 2633-entry covering list is a consistency check, not part of the trusted exhaustiveness argument.

8 Coding interpretation and the $q = 11$ application

This section is independent of the geometric proof of $\rho_{\mathcal{C}}(16) = 9$. It gives a second reading of the defect identity and then applies that dictionary to the $q = 11$ witness.

Choose nonzero column representatives of a k-arc $A \subset \mathrm{PG}(2,q)$ and form a $3 \times k$ matrix H_A. For $k \geq 4$, the code

$$C_A = \{c \in \mathbb{F}_q^k : H_A c^{\mathsf{T}} = 0\}$$

is a $[k, k-3, 4]_q$ maximum-distance-separable (MDS) code: every three columns are independent, the columns span $\mathbb{F}_q^3$, and any four columns support a dependence. Conversely, a projective parity-check system for a code with these parameters is a k-arc. This correspondence and its coset-weight consequences are classical; see Davydov–Marcugini–Pambianco [10].

We call such a code *projectively GRS* when, after permuting and scaling columns and applying a projective coordinate change, its projective parity-check columns lie on a normal rational curve. In $\mathrm{PG}(2,q)$ this is a nonsingular conic, by the standard normal-rational-curve/GRS correspondence [23]. The next proposition translates an uncovered point into a distance-three syndrome direction and a secant index into a minimum-leader count.

Proposition 8.1 (Syndrome form of relative completeness)**.** *Let $s \neq 0$ have projective direction $x = [s] \in \mathrm{PG}(2,q)$. Then:*

(i) the minimum weight of a word c with $H_A c^{\mathsf{T}} = s$ is one, two, or three according as $x \in A$, $x \notin A$ lies on a secant of A, or x is ordinarily uncovered;

(ii) if the minimum weight is two, the number of weight-two leaders is exactly $r_A(x)$;

(iii) if the minimum weight is three, the number of weight-three leaders is exactly $\binom{k}{3}$; and

(iv) adjoining x as one more projective parity-check column preserves the MDS property exactly when x is ordinarily uncovered.

Consequently, A is complete outside a prescribed set $\mathcal{H}$ exactly when all its projective distance-three syndrome directions lie in $\mathcal{H}$ (and $A \cap \mathcal{H} = \varnothing$). These directions are deep-hole directions only when covering radius three has separately been established.

Proof. A weight-one representation is a scalar multiple of a selected column. A weight-two representation uses a unique unordered column pair whose projective span is a secant through x; independence of the pair makes its two coefficients unique.

If neither case occurs, any three columns form a basis and represent s. For each three-subset the representation is unique, and none of its coefficients can vanish, so there are exactly $\binom{k}{3}$ weight-three leaders. The extension statement is the same three-column independence test. The leader count also follows as the $d = 4$ case of the general farthest-coset formula in Davydov–Marcugini–Pambianco [10, Theorem 7.7]; the deep-hole/MDS-extension formulation is standard in the Reed–Solomon setting [18]. □

Thus $r_A(x)$ is the exact number of minimum weight-two leaders of every affine syndrome on direction x. All $q - 1$ nonzero scalar multiples of a projective syndrome direction have the same distance and leader count: scalar multiplication bijects their coefficient words without changing Hamming support. This accounts for every projective-to-affine factor of $q - 1$ below.

The first moment counts leaders, the second counts collisions between distinct leader supports, and Theorem 3.1 is an exact leader-collision remainder after prescribed directions are removed. Thus Corollary 3.3 is also a length obstruction for projective codimension-three MDS codes whose distance-three syndrome directions are confined to the prescribed projective set.

8.1 The Clebsch witness over $\mathbb{F}_{11}$

The six-point witness over $\mathbb{F}_{11}$ is the classical Clebsch hexagon with icosahedral A_5 symmetry; see Edge, Dye, and Storme–Van Maldeghem for the configuration and its orbit geometry [9, 11, 22]. Dye's edge criterion and the complete-exterior formulation of Blokhuis–Seress–Wilbrink show that its secants avoid the conic, so $\mathcal{C}(\mathbb{F}_{11}) \subseteq U(A)$ [11, discussion preceding Theorem 6, p. 281] [8, pp. 143, 146]. The verified relative completeness of the witness gives the reverse inclusion and hence $U(A) = \mathcal{C}(\mathbb{F}_{11})$.

This equality is exceptional. For comparison, every four-arc has three repeated off-arc intersections among its six secants, and therefore

$$|U(A)| = (q^2 + q + 1 - 4) - \big(6(q-1) - 3\big) = (q-2)(q-3).$$

The exterior four-arc over $\mathbb{F}_7$ has $|U(A)| = 20 > 8 = |\mathcal{C}(\mathbb{F}_7)|$, so its conic lies strictly inside the uncovered locus [8, pp. 143, 146].

The verified incidence statistics place the $q = 11$ equality among the other exact examples:

q	$\lvert A\rvert$	$\binom{\lvert A\rvert}{2}$	$q^2 - \lvert A\rvert$	$I_{\mathcal{C}}(A)$	$\max_{x \notin A} r(x)$
8	6	15	58	16	3
9	6	15	75	13	3
11	6	15	115	0	3
16	9	36	247	32	4

In every row the last entry equals $m = \lfloor |A|/2 \rfloor$, one of the two extreme indices singled out by Corollary 3.4. The table does not assert small defect, but it records that the maximal-multiplicity part of the equality pattern is present in each exact witness.

The row at $q = 11$ has a stronger geometric interpretation. Since $I_{\mathcal{C}}(A) = 0$, none of the 15 secants meets $\mathcal{C}$: every secant of the witness is exterior to the conic. If N_i denotes the number of required points of index i, relative completeness and the maximum-index bound give only $i = 1, 2, 3$, while the two moment equations give

$$N_1 + N_2 + N_3 = 115, \qquad N_1 + 2N_2 + 3N_3 = 150, \qquad N_2 + 3N_3 = 45.$$

Consequently $(N_1, N_2, N_3) = (90, 15, 10)$.

Under the dictionary above, the equality $U(A) = \mathcal{C}(\mathbb{F}_{11})$ says that the associated code has precisely the conic as its projective distance-three syndrome locus. The next proposition records the resulting code and coset data.

Proposition 8.2 (The $q = 11$ code and coset spectrum)**.** *Let A be the six-point witness in Appendix C, and let C_A be its parity-check code. Then:*

(i) C_A is a projectively non-GRS $[6, 3, 4]_{11}$ MDS code of covering radius three, and its projective deep-hole syndrome locus is exactly the twelve-point conic $\mathcal{C}$;

(ii) the affine cosets of distances $0, 1, 2, 3$ number $(1, 60, 1150, 120)$, and the distance-two cosets split as $(900, 150, 100)$ according as they have one, two, or three minimum-weight leaders.

Proof. Every witness triple has nonzero determinant and the six columns span $\mathbb{F}_{11}^3$, giving the MDS parameters. The 6×6 evaluation matrix of the quadratic monomials

$$X^2,\ XY,\ XZ,\ Y^2,\ YZ,\ Z^2$$

is nonsingular, so no conic contains all six projective columns; by the preceding definition and the classical normal-rational-curve dictionary, the code is not projectively GRS. Direct finite arithmetic gives the secant-index spectrum

$$0 : 12, \qquad 1 : 90, \qquad 2 : 15, \qquad 3 : 10,$$

where index zero is precisely the conic. Multiplying each nonzero projective direction by its ten affine scalars gives all nonzero affine syndromes bijectively. Proposition 8.1 identifies the determinant-zero witness pairs with weight-two leader supports; uniqueness on an independent support and scalar multiplication then give the claimed coset and leader counts.

The twelve index-zero directions yield affine syndromes of distance three. Every syndrome has distance at most three because any three parity-check columns form a basis. Hence the covering radius is three, and these twelve directions are exactly the projective deep-hole locus. □

Optional extension structure. Relative completeness confines every further projective MDS extension to the twelve conic points. On those points, two extensions conflict exactly when their chord contains a witness point; the residual extension problem is therefore a finite graph-independence problem.

Proposition 8.3 (The $q = 11$ extension complex)**.** *For the same witness:*

(i) the simultaneous one-column MDS extensions form the independence complex of the icosahedral graph, with independence polynomial

$$1 + 12t + 36t^2 + 20t^3;$$

exactly six two-column and twenty three-column extensions are maximal, giving six complete eight-arcs and twenty complete nine-arcs containing the fixed seed, and there is no ten-arc extension;

(ii) each witness column colours five pairwise disjoint edges of the icosahedron, missing an antipodal pair, and the six colour classes partition its thirty edges. Adding the six antipodal edges turns them into six perfect matchings of the augmented graph.

Proof. For simultaneous extensions, two conic columns conflict exactly when their chord contains a witness column. No three conic columns are collinear, so there is no additional triple obstruction and valid extension sets are exactly graph-independent sets. Exhausting the 2^{12} subsets gives the displayed polynomial and maximal-set counts. The same determinant table partitions the thirty conflict edges into the six stated matchings. Thus maximality in the independence complex upgrades to ordinary completeness. The six missing antipodal edges are distinct; adjoining the appropriate one to each colour class gives six disjoint perfect matchings of the augmented graph. Appendix A records the two independent implementations and the formal checks. $\square$

Large-field reconstruction from the uncovered locus. At every fixed length, the complete set of one-column MDS extensions eventually remembers the projective parity-check system itself. The inverse map has two elementary stages. Put $N = \binom{k}{2}$ and $S = \mathrm{PG}(2, q) \setminus U(A)$. First recover the secant arrangement from the gap between $q + 1$, the number of points on a genuine secant, and N, the largest number of points in which a nonsecant line can meet the union. Then recover the vertices from the gap between their secant multiplicity $k - 1$ and the largest possible nonvertex multiplicity $\lfloor k/2 \rfloor$:

$$U(A) \longmapsto S \longmapsto \{\ell : |\ell \cap S| > N\} \longmapsto \{x : \deg(x) = k - 1\} = A.$$

The strict first gap is the only large-field hypothesis.

Proposition 8.4 (Uncovered-locus reconstruction)**.** *Let A and B be k-arcs in* $\mathrm{PG}(2, q)$*, where $k \geq 3$ and*

$$q + 1 > \binom{k}{2}.$$

If $U(A) = U(B)$ as literal subsets of the plane, then $A = B$ as unlabelled point sets.

Proof. The complement of $U(A)$ is the union of the $\binom{k}{2}$ distinct secants of A, and likewise for B. Let ℓ be a secant of A. If no secant of B were equal to ℓ, then the secants of B would cover at most $\binom{k}{2}$ points of ℓ, one per line. This is impossible because $|\ell| = q + 1 > \binom{k}{2}$ and the two secant unions are equal. Thus every secant of A is a secant of B; since the two sets have the same finite size, their secant-line sets coincide.

It remains to recover the vertices from this line arrangement. Each point of A lies on its $k - 1$ incident secants. A point outside A lies on at most $\lfloor k/2 \rfloor$ secants: distinct secants through it use disjoint pairs of the k vertices and hence form a matching. Since $k - 1 > \lfloor k/2 \rfloor$ for $k \geq 3$, the set A is exactly the set of points incident with $k - 1$ recovered secants, and the same description recovers B. $\square$

Corollary 8.5 (Canonical inverse and automorphisms). *Under the hypotheses of Proposition 8.4, put $S = \mathrm{PG}(2, q) \setminus U(A)$ and $N = \binom{k}{2}$. The secants of A are exactly the lines ℓ with $|\ell \cap S| > N$, and A is exactly the set of points incident with $k-1$ such lines. Moreover,*

$$\mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(U(A)) = \mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(A).$$

Proof. Every secant lies in S and has $q + 1 > N$ points. A nonsecant line meets each of the N secants in at most one point, so it contains at most N points of their union S. The vertex-recovery rule is the one proved above. Finally, every semilinear projectivity stabilizing A stabilizes $U(A)$. Conversely, if $gU(A) = U(A)$, then $U(gA) = gU(A) = U(A)$, so Proposition 8.4 gives $gA = A$. □

Remark 8.6 (Sharpness of the strict threshold). The strict inequality cannot in general be weakened to $q + 1 \geq \binom{k}{2}$. For the projective four-frame A over $\mathbb{F}_5$, Proposition 7.2 gives $U(A) = \mathcal{C}(\mathbb{F}_5)$, while $q + 1 = \binom{4}{2} = 6$. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}_3(5)$ has order $|\mathrm{PGL}_2(5)| = 120$, whereas the setwise stabilizer of a projective frame has order $4! = 24$. The conic stabilizer therefore moves A through at least five distinct four-frames, all with the same literal uncovered locus $\mathcal{C}(\mathbb{F}_5)$.

9 Conclusion and further questions

The exact defect identity is the organizing result of the paper. It separates the universal overlap forced by the second secant moment from the incidence spent on a prescribed exceptional set, and its nonnegative local terms retain the equality and stability information lost by a scalar counting bound. Its line-hole specialization recovers complete affine arcs, while the conic case supplies a curved exceptional locus. The coding interpretation is another reading of the same remainder, rather than a separate source of estimates.

The forward defect identity records how a secant arrangement covers the plane; the reconstruction theorem supplies a partial inverse. The uncovered locus is therefore more than an output statistic: Proposition 8.4 recovers the unlabelled parity-check arc from its complete set of distance-three syndrome directions whenever $q + 1 > \binom{k}{2}$. Thus the one-column extension locus itself becomes a parent fingerprint once every secant contains more points than there are competing secants; for six columns this applies when $q \geq 16$. Corollary 8.5 makes the recovery canonical and shows that the same locus determines the full projective-semilinear automorphism group. Remark 8.6 shows that the strict incidence threshold is best possible in general.

Two further mechanisms complement that identity. Projective averaging transfers ordinary complete arcs to prescribed-hole upper bounds, while the evaluation-rank obstruction detects finite configurations invisible to the incidence budget. Over $\mathbb{F}_{16}$ the defect bound reaches eight and the quadratic obstruction excludes eight, leaving the checked nine-point witness and hence $\rho_{\mathcal{C}}(16) = 9$.

These mechanisms leave four concrete directions.

Problem 9.1. Determine whether the uncovered-quadratic-rank obstruction used at $q = 16$ extends to other even orders or gives useful lower bounds beyond the scalar defect inequality.

Problem 9.2. Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 5.1 currently supplies only the general complete-arc bound $O(\sqrt{q}(\log q)^c)$.

Problem 9.3. Classify equality and bounded-defect configurations in Theorem 3.1, and determine whether the conic stabilizer yields finite-order restrictions not reducible to a bound on $I_{\mathcal{C}}(A)$. Remark 4.4 shows that such a bound alone cannot improve the asymptotic constant obtained here.

Problem 9.4. Develop the higher-dimensional analogue for caps complete outside a prescribed quadric in $\mathrm{PG}(n, q)$. The evaluation lemma already applies to arbitrary feature maps and Veronese degrees; the missing ingredient is an appropriate exact secant-moment remainder for caps.

Progress beyond the present lower bound will probably require an explicit construction, an algebraic obstruction, or a classification of near-extremal secant arrangements rather than another application of the same incidence budget.

A Computational and formal verification

The body separates mathematical reductions from finite exclusion. The corresponding evidence and trust boundaries are summarized below. The source supplement's `TRUST.md` manifest records exact commands, toolchain versions, SHA-256 digests, theorem names, and frozen outputs.

Claim	Evidence	Trust boundary
Five upper-bound witnesses	The dedicated q=5 Lean leaf checks the transported frame; the general verifier and Lean leaves check q=8,9,11,16	Every displayed coordinate list has a rules-only Lean certificate; the independent Python verifier covers q=8,9,11,16 only; no search-completeness claim
General theorem chain	Companion Lean library formalizes the incidence, defect, conic, asymptotic, averaging, nucleus, evaluation, syndrome, and coding arguments	Kernel checking with axioms `propext`, `Classical.choice`, and `Quot.sound`; no `sorry`, custom axiom, or `native_decide`
$q = 16$ classification	C++ generator emits local transition and quadratic-evaluation data for covering lists of lengths 4/61/454/2633	Lean checks transition coverage, leaf rejection, the exact $2630 + 3$ profile, the three exceptional kernel lines, and the full arbitrary-eight-arc quadratic-avoidance theorem, including coordinate pullback, preservation of nonzeroness, and $U(gA) = gU(A)$. Generator assertions, canonical labels, and deduplication are not trusted
$q = 11$ code and extension data	Lean checks the stated code, syndrome, graph claims, and the implication from the classical NRC/GRS quadratic consequence to non-GRS; independently written Python and C++ programs regenerate the finite data and include coordinate-invariance and mutation controls	The NRC/GRS dictionary remains the cited classical input; the two programs are independent reproducibility checks

For the additive lower bound, the formal theorem contains the same cubic expansion and the same coefficient estimates isolated in the body's elementary estimate. Along unbounded families of finite-field orders, Lean proves unconditionally that every number below 3/2 is eventually a lower bound for the centred parameter. Its literal real-valued `liminf` wrapper assumes the centred family is cobounded, because a real-valued `liminf` does not represent divergence to $+\infty$; the operational statement covers that case as well.

The q=5 witness entry point is the dedicated rules-only Lean leaf `RelativeConicArcs/ExampleChecks/Q5.lean`; the C187 checker independently replays the frame's six-point uncovered-conic identity, while Lean also checks the displayed invertible coordinate matrix and pullback identity. For q=8,9,11,16, the independent witness entry point is `verify_relative_conic_arcs.py`, and the corresponding Lean leaves check the displayed coordinates. All five Lean certificates verify disjointness, the arc condition, and coverage of the canonical projective representatives $[1 : y : z]$, $[0 : 1 : z]$, and $[0 : 0 : 1]$; a soundness theorem transports acceptance to relative completeness. The q=16 classification entry point is `search_rhoc16.cpp`; its checked-in report and emitted Lean data fix the finite search input. The finite examples do not depend on the external Kim–Vu hypothesis used only for the conditional asymptotic upper bound.

The three exceptional leaves. Using the binary $\mathbb{F}_{16}$ encoding from Appendix C and coefficient order $(X^2, Y^2, Z^2, XY, XZ, YZ)$, the one-dimensional kernels of the three exceptional evaluation matrices are generated by

$$(1, 1, 1, 1, 1, 0), \qquad (0, 0, 0, 5, 4, 1), \qquad (2, 1, 1, 5, 5, 1).$$

Lean identifies these with the three forced-hit records in the exhaustive list and proves that each displayed line is the exact kernel. The first and third forms split over $\mathbb{F}_{16}$ as

$$(X + 6Y + 6Z)(X + 7Y + 7Z) \quad \text{and} \quad 2(X + 4Y + 15Z)(X + 15Y + 4Z),$$

and each union of lines contains two selected points. The middle form is nonsingular and contains seven selected points. Thus two rank defects come from reducible quadratics, while the third comes from an arc differing by one point from a seven-point conic subset. These descriptions are not used for classification completeness.

B A general finite-field evaluation dichotomy

The one-selected-point obstruction used in Section 7 has a sharp extension to several selected points. The vector-space covering threshold is classical; see, for example, Khare [19].

Proposition B.1 (Finite-field evaluation dichotomy)**.** *Let W be a finite-dimensional vector space over $\mathbb{F}_q$, let $\nu : X \to W$ be any feature map, and let $U, A \subseteq X$ be finite with $|A| \le q$. There is a linear form $f \in W^*$ such that*

$$f(\nu(u)) = 0 \quad (u \in U), \qquad f \neq 0, \qquad f(\nu(a)) \neq 0 \quad (a \in A)$$

if and only if

$$\langle \nu(U) \rangle \neq W \quad \text{and} \quad \nu(a) \notin \langle \nu(U) \rangle \quad (a \in A).$$

The uniform threshold is sharp: $q + 1$ distinct hyperplanes through a common codimension-two subspace can cover W.

Proof. Put $S = \langle \nu(U) \rangle$. Necessity follows by applying f to the stated span relations. Conversely, $S \neq W$ makes the annihilator $S^\perp \subseteq W^*$ nonzero. For each $a \in A$, the condition $\nu(a) \notin S$ says that evaluation at $\nu(a)$ cuts out a proper hyperplane of $S^\perp$. At most q

proper hyperplanes cannot cover a nonzero vector space over $\mathbb{F}_q$: their union has size at most $1 + q(|S^\perp|/q - 1) < |S^\perp|$. Any form outside their union has the required properties. For sharpness, quotient by a common codimension-two subspace and use all $q+1$ lines in $\mathbb{F}_q^2$. □

For the degree-d Veronese map, the proposition gives the same dichotomy for homogeneous degree-d forms.

C Witness coordinates and field encodings

All coordinates are homogeneous and use the standard conic $XZ = Y^2$. The rules-only Lean checkers in `RelativeConicArcs/ExampleChecks/`—namely `Q5.lean`, `Q8.lean`, `Q9.lean`, `Q11.lean`, and `Q16.lean`—verify the displayed coordinates and feed the exact-value theorems in the aggregate `Results.lean`. The independent Python entry point `verify_relative_conic_arcs.py` regenerates the q=8,9,11,16 witness checks but is not part of the Lean trust base. The q=5 frame identity has the separate C187 checker described above.

For $q = 5$, coordinates lie in the prime field. The projective image of the standard four-frame used by the certificate is

$$(1, 0, 3), (2, 1, 2), (3, 3, 4), (1, 4, 4).$$

It is obtained from the frame by the matrix with rows $(1, 2, 3)$, $(0, 1, 3)$, $(3, 2, 4)$, whose determinant is 2; substitution carries the displayed frame conic to a nonzero multiple of $XZ - Y^2$.

For $q = 8$, write $\mathbb{F}_8 = \mathbb{F}_2[\alpha]/(\alpha^3 + \alpha + 1)$ and encode $a_0 + a_1\alpha + a_2\alpha^2$ by the binary integer $a_0 + 2a_1 + 4a_2$. A witness is

$$(0, 1, 1), (0, 1, 2), (1, 0, 1),$$
$$(1, 0, 2), (1, 1, 2), (1, 1, 6).$$

For $q = 9$, write $\mathbb{F}_9 = \mathbb{F}_3[\alpha]/(\alpha^2 + 1)$ and encode $a_0 + a_1\alpha$ by $a_0 + 3a_1$. A witness is

$$(1, 0, 4), (1, 0, 5), (1, 1, 0),$$
$$(1, 1, 2), (1, 2, 3), (1, 2, 4).$$

For $q = 11$, coordinates lie in the prime field. A witness is

$$(1, 10, 0), (1, 9, 1), (1, 4, 7),$$
$$(1, 8, 5), (0, 1, 4), (1, 1, 7).$$

For $q = 16$, write $\mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$ and use binary coefficient encoding. A witness of size nine is

$$(1, 5, 14), (1, 5, 3), (1, 9, 2), (1, 0, 9), (1, 13, 13),$$
$$(1, 6, 10), (1, 3, 9), (0, 1, 11), (1, 10, 2).$$

References

[1] T. Rudd, The Clebsch hexagon code: rigidity from a conic deep-hole locus, companion manuscript, 2026.

[2] S. Alabdullah and J. W. P. Hirschfeld, A new lower bound for the smallest complete (k,n)-arc in $\mathrm{PG}(2,q)$, *Designs, Codes and Cryptography* **87** (2019), 679–683. doi:10.1007/s10623-018-00592-8.

[3] N. A. M. Al-Seraji and R. A. B. Al-Ogali, Classification of arcs in finite projective plane of order sixteen, *Al-Mustansiriyah Journal of Science* **29**(1) (2018), 138–149. doi:10.23851/mjs.v29i1.184.

[4] B. Alspach and K. Heinrich, Matching designs, *Australasian Journal of Combinatorics* **2** (1990), 39–55.

[5] S. Ball, On small complete arcs in a finite plane, *Discrete Mathematics* **174** (1997), 29–34. doi:10.1016/S0012-365X(96)00315-9.

[6] S. Ball, On arcs and quadrics, in *Arithmetic of Finite Fields—WAIFI 2016*, Lecture Notes in Computer Science **10064**, Springer, 2017, 95–102. doi:10.1007/978-3-319-55227-9_7.

[7] D. Bartoli, A. A. Davydov, S. Marcugini, and F. Pambianco, On the smallest size of an almost complete subset of a conic in $\mathrm{PG}(2,q)$ and extendability of Reed–Solomon codes, *Problems of Information Transmission* **54** (2018), 101–115. doi:10.1134/S0032946018020011.

[8] A. Blokhuis, A. Seress, and H. A. Wilbrink, Characterization of complete exterior sets of conics, *Combinatorica* **12**(2) (1992), 143–147. doi:10.1007/BF01204717.

[9] W. L. Edge, Conics and orthogonal projectivities in a finite plane, *Canadian Journal of Mathematics* **8** (1956), 362–382.

[10] A. A. Davydov, S. Marcugini, and F. Pambianco, On the weight distribution of the cosets of MDS codes, *Advances in Mathematics of Communications* **17**(5) (2023), 1115–1138. doi:10.3934/amc.2021042. arXiv:2101.12722v2. (Numbered results are cited by their arXiv v2 numbering.)

[11] R. H. Dye, Hexagons, conics, A_5 and $\mathrm{PSL}_2(K)$, *Journal of the London Mathematical Society* (2) **44** (1991), 270–286. doi:10.1112/jlms/s2-44.2.270.

[12] D. G. Glynn, On the construction of arcs using quadrics, *Australasian Journal of Combinatorics* **9** (1994), 3–19.

[13] M. Giulietti and E. Montanucci, On hyperfocused arcs in $\mathrm{PG}(2,q)$, *Discrete Mathematics* **306**(24) (2006), 3307–3314. doi:10.1016/j.disc.2006.06.009.

[14] J. W. P. Hirschfeld, *Projective Geometries over Finite Fields*, second edition, Oxford University Press, 1998.

[15] J. H. Kim and V. H. Vu, Small complete arcs in projective planes, *Combinatorica* **23** (2003), 311–363. doi:10.1007/s00493-003-0024-1.

[16] G. Korchmáros and T. Szőnyi, Affinely regular polygons in an affine plane, *Contributions to Discrete Mathematics* **3**(1) (2008), 20–38. doi:10.55016/ojs/cdm.v3i1.62767.

[17] G. Korchmáros, G. P. Nagy, and T. Szőnyi, Algebraic approach to the completeness problem for (k,n)-arcs in planes over finite fields, *Journal of Combinatorial Theory, Series A* **204** (2024), 105851. doi:10.1016/j.jcta.2023.105851. arXiv:2302.10162.

[18] K. Kaipa, Deep holes and MDS extensions of Reed–Solomon codes, *IEEE Transactions on Information Theory* **63** (2017), 4940–4948.

[19] A. Khare, Vector spaces as unions of proper subspaces, *Linear Algebra and its Applications* **431** (2009), 1681–1686. doi:10.1016/j.laa.2009.06.001.

[20] Z. L. Nagy, Saturating sets in projective planes and hypergraph covers, *Discrete Mathematics* **341** (2018), 1078–1083. doi:10.1016/j.disc.2018.01.011.

[21] S.-L. Ng and P. R. Wild, On k-arcs covering a line in finite projective planes, *Ars Combinatoria* **58** (2001), 289–300.

[22] L. Storme and H. Van Maldeghem, Primitive arcs in $\mathrm{PG}(2, q)$, *Journal of Combinatorial Theory, Series A* **69** (1995), 200–216. doi:10.1016/0097-3165(95)90051-9.

[23] L. Storme and J. A. Thas, Generalized Reed–Solomon codes and normal rational curves: an improvement of results by Seroussi and Roth, in *Advances in Finite Geometries and Designs*, Oxford University Press, 1991, 369–389. doi:10.1093/oso/9780198535928.003.0031.